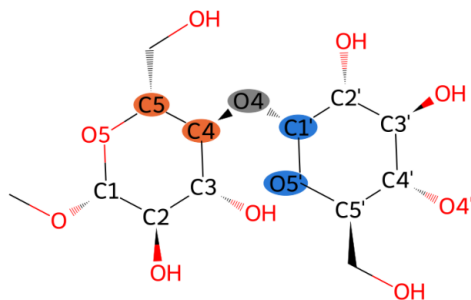
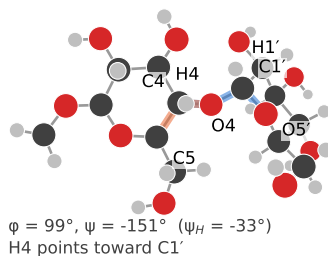


a the $\alpha(1\rightarrow4)$ linkage of amylose (methyl α -maltoside): $\phi = \text{O5}'\text{-C1}'\text{-O4-C4}$ (blue), $\psi = \text{C1}'\text{-O4-C4-C5}$ (orange)

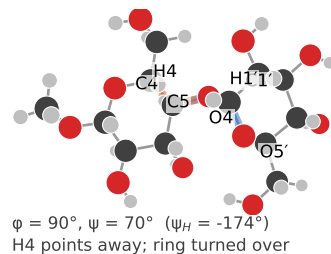


unprimed ring: reducing side (carries C4 and H4); primed ring: non-reducing side (carries C1'). Hydrogen convention of Kuttel-Naidoo and Hatcher: $\phi_H = \text{H1}'\text{-C1}'\text{-O4-C4}$, $\psi_H = \text{C1}'\text{-O4-C4-H4}$ (here $\phi = \phi_H + 116^\circ$, $\psi = \psi_H - 118^\circ$). The next residue starts at O4'; the chain of O4 atoms is the virtual chain.

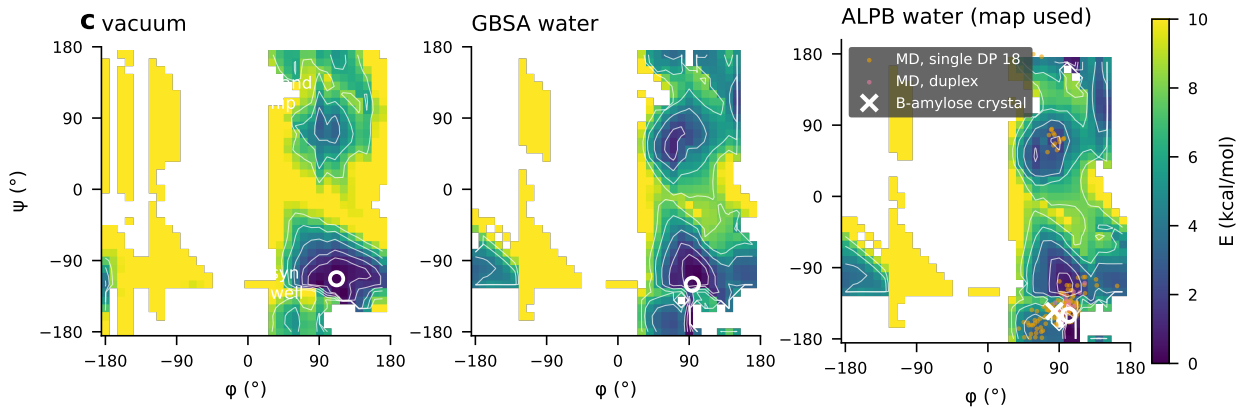
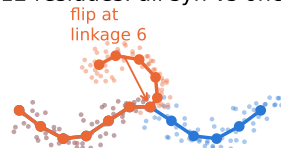
b syn ($100^\circ, -150^\circ$)



band flip ($90^\circ, 70^\circ$)



12 residues: all syn vs one flip



c three GFN2-xTB maps of the same linkage. Circle: Boltzmann mean of the syn population at 303 K; it slides along the flat valley with solvation. The band-flip well lies 2.9 (vacuum), 1.1 (GBSA) and 1.7 (ALPB) kcal/mol above the syn minimum; white = rejected cells (ring deformed).